

# Route Finding App

Dropped as a Table

# Our Problem

- People would often rather take a slightly longer route than have to walk on a route with lots of crossings or a lack of greenery etc.
- However, mainstream route finding apps only give users the most time efficient route and do not take into account other indicators
- There is a need for an app, specifically for walkers that allows users to set their preferences and generate routes based on those preferences



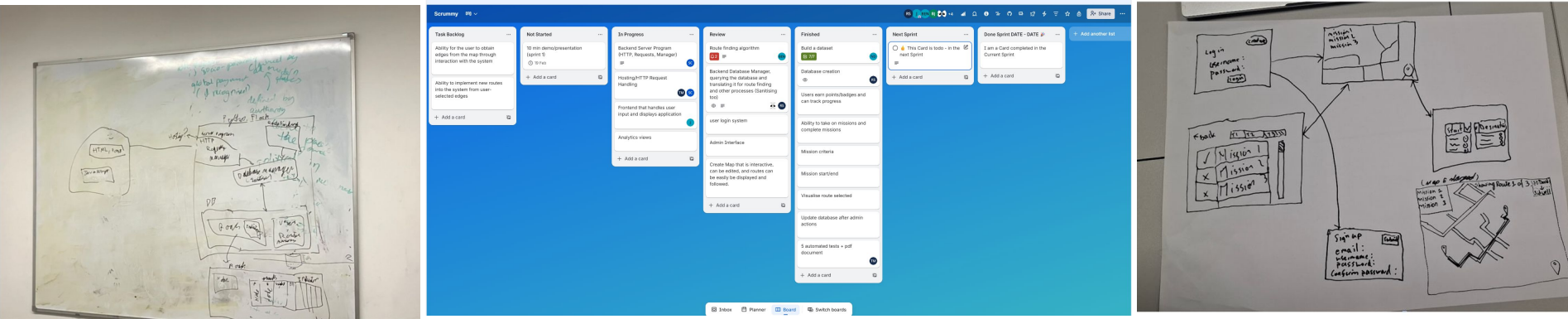
# Approach and Design

For this project, we took an Agile approach, where we had weekly sprints and scrum meetings.

We broke down the project into smaller tasks (displayed on the scrum board pictured below) and during the scrum meetings we updated each other on progress and took on new tasks for the next week long sprint.

We all also took on a role which we were responsible for (i.e. Databases and SQL, Testing, Programming).

This was also where we decided the overall structure of the project and how the various states (map screen, mission select, etc...) would lead into each other.



# Implementation

Created a database From Openstreetmap Data and managed it with sqlite

Used Flask, HTML, CSS, and JavaScript to create the web app

Used SciPy for the route finding algorithm between locations

| nodeID | coordinatesX | coordinatesY | lighting | crime | greenery | gradient |
|--------|--------------|--------------|----------|-------|----------|----------|
| 0      | -3.511056    | 50.722296    | 0.37     | 0.8   | 0.73     | 0.6      |
| 1      | -3.535938    | 50.718241    | 0.16     | 0.01  | 0.06     | 0.87     |
| 2      | -3.536843    | 50.71763     | 0.6      | 0.56  | 0.02     | 0.97     |
| 3      | -3.537705    | 50.717846    | 0.83     | 0.06  | 0.18     | 0.18     |
| 4      | -3.538113    | 50.718031    | 0.3      | 0.37  | 0.43     | 0.29     |
| 5      | -3.536987    | 50.718869    | 0.61     | 0     | 0.29     | 0.37     |
| 6      | -3.536861    | 50.719546    | 0.46     | 0.64  | 0.2      | 0.51     |
| 7      | -3.535776    | 50.719558    | 0.59     | 0     | 0.61     | 0.17     |
| 8      | -3.533024    | 50.719619    | 0.07     | 0.8   | 0.97     | 0.81     |
| 9      | -3.529419    | 50.719905    | 0.3      | 0     | 0.68     | 0.44     |
| 10     | -3.522206    | 50.725268    | 0.12     | 0.35  | 0.03     | 0.91     |
| 11     | -3.523003    | 50.72532     | 0.26     | 0.51  | 0.31     | 0.52     |
| 12     | -3.523353    | 50.725198    | 0.55     | 0.03  | 0.97     | 0.78     |
| 13     | -3.508812    | 50.721817    | 0.94     | 0.74  | 0.6      | 0.92     |
| 14     | -3.513665    | 50.722946    | 0.09     | 0.05  | 0.05     | 0.33     |
| 15     | -3.516527    | 50.723639    | 0.39     | 0.12  | 0.83     | 0.36     |
| 16     | -3.523108    | 50.727661    | 0.28     | 0.39  | 0.14     | 0.8      |
| 17     | -3.53639     | 50.717694    | 0.07     | 0.84  | 0.77     | 0.2      |
| 18     | -3.541471    | 50.719539    | 0.01     | 0.67  | 0.71     | 0.73     |
| 19     | -3.538085    | 50.718403    | 0.77     | 0     | 0.36     | 0.12     |
| 20     | -3.538697    | 50.718332    | 0.86     | 0.47  | 0.33     | 0.06     |
| 21     | -3.541381    | 50.732525    | 0.31     | 0.18  | 0.73     | 0.64     |
| 22     | -3.535913    | 50.727577    | 0.89     | 0.32  | 0.12     | 0.71     |

Custom Routes in Exeter, UK  
This tool does not reflect real world data  
This tool is not suitable for emergency use

Route Finder

Start Node:  End Node:  (Refresh) (Clear)

Start Location:  End Node:  (Refresh) (Clear) Distance:  Lighting:  Greenery:  Gradient:  Crime:

Route from Node 263 to Job:

Red Path: 263 → 262 → 273 → 654 → 392 → 393 → 451 → 282 → 419 → 204 → 223 → 785 → 355 → 775 → 360

Total Distance: 772.00

Score Breakdown: Lighting: 0.51 Crime: 0.34 Greenery: 0.52 Gradient: 0.34

Green Path: 263 → 262 → 272 → 170 → 546 → 603 → 259 → 714 → 363 → 366 → 451 → 282 → 419 → 204 → 223 → 785 → 355 → 775 → 360

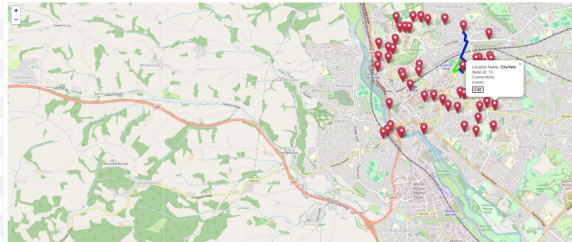
Total Distance: 812.00

Score Breakdown: Lighting: 0.51 Crime: 0.29 Greenery: 0.50 Gradient: 0.49

Blue Path: 263 → 262 → 273 → 654 → 392 → 393 → 451 → 282 → 419 → 204 → 223 → 785 → 355 → 775 → 360

Total Distance: 772.00

Score Breakdown: Lighting: 0.51 Crime: 0.34 Greenery: 0.52 Gradient: 0.34



# Demo

- We will show how the app finds multiple routes and how it made that decision.
- We will show that missions can be completed with feedback given to users.
- We will show that nodes can be added to the map by a trusted user.
- We will show that missions can be edited by trusted users.

# Limitations

GUI and UX - 'Prettify' the main display page.

Execution time - Improve the route-finding speed.

Low variance in routes - Currently there is much overlap in routes.



# Sprint 2

## Requirements

- Have at least 6 viewable analytics (e.g. most popular destination) across the system.
- Audit log to track edits made.
- Role based access control (login, session management).
- Mobile-friendly layout.
- User feedback.

## Risks

- Scope creep.
- Straying from project requirements.
- Problems from Sprint 1 flowing into Sprint 2.